# EJIGA PETER OJONUGWA OLUWAFEMI

**Machine Learning Research Engineer | Technical AI Safety | Multimodal and Foundation Models**

Reading, Pennsylvania, USA | +1 443 712 3126 | ejiga.ojonugwa.peter@gmail.com

linkedin.com/in/ejigapeter | github.com/ejigsonpeter | scholar.google.com/citations?user=F5b2esYAAAAJ | ORCID: 0009-0003-2039-3075

## PROFESSIONAL SUMMARY

Machine learning research engineer with 7+ years across software engineering, research, and technical development, including formal research-lab experience since 2023. Hands-on experience with LLM-agent safety evaluation, RLHF-oriented model evaluation, multimodal foundation models, generative AI, model fine-tuning, and reproducible ML experimentation. Current work studies when long-horizon LLM-agent failures become observable and how monitoring thresholds transfer across model families. Experienced in Python, PyTorch, Hugging Face, AWS, model evaluation, and translating research prototypes into usable systems.

## CORE TECHNICAL SKILLS

**AI Safety:** LLM agent evaluation, runtime monitoring, failure analysis, model robustness, alignment and misalignment concepts, model forensics, safety evaluation, interpretability concepts (BlueDot training)

**Models and Training:** Qwen2.5-VL, Florence-2, Stable Diffusion, DreamBooth, LoRA, QLoRA, PEFT, vision-language models, transformers, computer vision, NLP

**Engineering and Evaluation:** Python, PyTorch, Hugging Face, scikit-learn, pandas, Git, REST/API workflows, reproducible pipelines, ablation studies, statistical evaluation, AWS, Azure ML

## SELECTED AI SAFETY AND ML RESEARCH

**When Do Agent Failures Become Visible? Model-Dependent Warning Horizons in LLM Agents** | Lead Researcher, manuscript in preparation | 2026

- Built a reproducible long-horizon LLM-agent evaluation pipeline on tau-bench retail spanning GPT-4o-mini, Gemini Flash, Qwen3-Max, Grok-4, and Claude Haiku 4.5.
- Analyzed 679 real agent rollouts and three trajectory-divergence signals to determine when impending failure becomes observable from partial trajectories.
- Found model-dependent warning horizons and early-onset, late-onset, and judge-illegible failure phenotypes; pooled early-warning signal reached AUROC 0.73 at step 5.
- Designed a frozen, outcome-blind annotation pipeline so model rollouts are generated once and all downstream safety analyses are reproducible from stored logs.
- Reported and diagnosed a null result for perturbation-amplification rather than forcing a positive finding, testing metric saturation, perturbation placement, and task headroom.

**RLHF and Model Evaluation Contributor, Project HedgeDog** | Handshake | Remote | April 2026 - July 2026

- Completed RLHF-oriented model evaluation tasks using structured rubrics and project guidelines to assess response quality, preference alignment, instruction following, and model behavior.
- Reviewed model outputs for quality and consistency, contributing human evaluation signals used in model-improvement workflows.

**Technical AI Safety Apprentice** | BlueDot Impact | Remote | March 2026 - April 2026

- Completed technical AI safety training covering AI misalignment, mechanistic explainability, frontier-model safety frameworks, dangerous-capability detection, and approaches to training safer models.

## RESEARCH AND ENGINEERING EXPERIENCE

**Graduate Research Assistant, Computer Vision and AI Lab** | Morgan State University | Baltimore, MD | 2023 - Present

- Conduct research in multimodal AI, generative modeling, medical imaging, model evaluation, and robustness; build reproducible PyTorch and Hugging Face experimentation pipelines.

- Fine-tuned Stable Diffusion and DreamBooth-based generative models for domain-specific text-to-image synthesis; used parameter-efficient adaptation including LoRA in research workflows.
- Fine-tuned and evaluated vision-language models including Qwen2.5-VL and Florence-2 for multimodal question answering and captioning tasks.
- Developed systems evaluated through CLEF organizer-run benchmarks, providing externally measured performance beyond internal train/validation splits.
- Contributed to peer-reviewed research and NSF/NIH-supported projects; analyze failure modes, distribution shift, calibration, and model generalization.

**AI Software Engineer (Part-Time)** | Dathway AI | Remote, USA | October 2024 - Present

- Develop Python and NLP-based AI systems and scalable workflows supporting model training, evaluation, and production deployment.
- Implement secure AI deployment practices aligned with DevSecOps and OWASP principles and mentor technical teams on responsible AI integration.

**Cloud Engineer** | Cloud Evo LLC | Remote, USA | August 2022 - August 2023

- Designed AWS cloud infrastructure and deployment workflows using services including CloudFormation, S3, Elastic Beanstalk, Aurora, and CloudFront; improved operational efficiency and security.
- Worked across backend systems and DevOps workflows using Node.js, Express, PostgreSQL, Git, and team-based software delivery.

**Software Engineer** | Archaeotech Solutions Ltd | Nigeria / Remote | January 2019 - July 2022

- Led software development for client applications using PHP, MySQL, JavaScript, and web technologies; worked across requirements, implementation, testing, and delivery.
- Conducted vulnerability assessments with Kali Linux and integrated security practices into application-development workflows.

## SELECTED PUBLICATIONS AND RESEARCH OUTPUTS

- Ejiga Peter Ojonugwa. "When Do Agent Failures Become Visible? Model-Dependent Warning Horizons in LLM Agents." Manuscript in preparation, 2026.
- "Text-Guided Synthesis in Medical Multimedia Retrieval: A Framework for Enhanced Colonoscopy Image Classification and Segmentation." Algorithms, 2025.
- "Advancing AI-Powered Medical Image Synthesis: Insights from MedVQA-GI Challenge Using CLIP, Fine-Tuned Stable Diffusion, and DreamBooth + LoRA." CLEF/CEUR, 2024.
- "Comparative Analysis of Fine-tuned Multimodal Models in Radiology Image Captioning." 2025.
- "Explainable MRI-Based Ensemble Learnable Architecture for Alzheimer's Disease Detection." 2025.

## EDUCATION

**M.S., Advanced Computing (Computer Science)** | Morgan State University | Baltimore, MD | 2024

**B.Tech., Computer Science** | Federal University of Technology, Minna | 2019

## CERTIFICATIONS AND TRAINING

Google DeepMind AGI Safety Course, Google DeepMind Safety Research (2026); Technical AI Safety Certification, BlueDot Impact (2026); CompTIA Security+ (2026); AWS Certified AI Practitioner (2025); AWS Certified Cloud Practitioner (2025); Huawei Certified Network Associate (2019)

## AWARDS AND PROFESSIONAL SERVICE

**Peer Review:** Reviewer for Elsevier Engineering Applications of Artificial Intelligence, Elsevier Image and Vision Computing, and IEEE BHI.

**Selected Awards:** SCMNS Graduate Student Award (2024); AWS AI and ML Scholarship Recipient (2024); Intel AI Scholarship Recipient (2019).